

CS 252a Syllabus

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Adapted (heavily) from slides by Adam Bates, Spring 2020

Learning Objectives

- Before CS 252a:
 - Intermediate knowledge of computer architecture topics
 - Experience working independently on machine problems involving systems programming, software engineering, networking, etc.
- After CS 252a:
 - Expert understanding of several advanced computer architecture topics
 - Able to engage and critique academic security literature
 - Able to effectively communicate architecture research in presentations
 - Able to independently conduct computer architecture research...
 - Identify interesting and novel research questions
 - Design study methodologies to answer these questions
 - Evaluate and analyze your results
 - Convey the importance of your findings to a broad audience

Course logistics

- **<https://docs.google.com/spreadsheets/d/1POjR4ilyCHdwYeaznIxF4VrJgSt-B7i3/edit?gid=1301587908#gid=1301587908>**

Class logistics

- Meet every week (except Spring break)
- Attending live is important
 - This course is discussion-centric
- If you cannot make lecture in person for an extended period of time
 - Let me know – we will try to accommodate

Course requirements

1. Read Literature: Read the 2 assigned papers in advance of each class
2. Participate: Come to class, W 5pm. Contribute to all class discussions.
3. Write Reaction Reviews: Prepare one “peer review”-style summary per paper per week.
4. Some one-off assignments (project idea brainstorming)
5. Present a paper: Present research papers and lead the ensuing class discussion. Everyone will present at least once.
6. Term Project: Conduct a major research project in architecture, with the chief deliverable being a conference-style paper at the end of the semester.

1 – Read Literature

- This semester, we will be reading and discussing important work from two broad areas and a potpourri of timely topics
- Broad area 1: Intellectually delightful / Foundational / Classics
- Broad area 2: Late-breaking papers
- All papers linked on the website

2 - Participate

- Attend class live and regularly contribute to discussion with questions/comments
- No electronics!
- How to do discussion
 - Speak up! Interrupt away.
 - We will have a class slack. Use it to post questions/comments/etc.

3 – Write reaction papers

- We'll be writing reaction papers in a “Peer Review”-style format in which you summarize, critique, and advocate for the assigned readings
- Potential outcome: You learn to think like a reviewer → *You will graduate sooner*
- Other reasons:
 - Reaction papers are a good accountability mechanism
 - Coming to class prepared shows respect to your peers that are presenting the paper
 - Critically engaging with literature is an important part of a career in research

Publishing reviews

- Before each class, I will make all reviews public
- Reviews will be due a few days before the presentation so that presenter has time to go through them

Review grading

- Reviews will be graded
 - 3 = outstanding review (we will call these out)
 - 2 = acceptable
 - 1 = needs improvement
 - 0 = unacceptable
- Reviews will be made public (to other students in the class)

Example reaction paper/review #1

Paper: [Hassan et al., USENIX Security 2018]

Overall Merit: Accept

Summary: This paper presents a data-driven study of the privacy issues of fitness tracking social networks. The authors collect 21M posts over a period of one month from the Strava fitness social networks, and investigate the use of privacy measures such as Endpoint Privacy Zones (EPZs) by its users. They then go on to develop an attack that leverages multiple data points to infer the private location (usually the user's home), showing that the attack is successful. They then evaluate two state of the art privacy mechanisms (modify radius size and spatial cloaking) and design a new one (fuzz EPZ intersection points) to prevent this attack, showing that although these methods are helpful, a motivated attacker could still defeat them. The authors performed responsible disclosure to Strava, who is now in the process of implementing geo-indistinguishability techniques on their platform.

Strengths:

- Well executed data-driven study taking advantage of real-world data and looking at multiple platforms.
- The paper both tests existing privacy-preserving measures and develops a new one to limit the problem.
- The research had real world impact since fitness tracking companies are developing countermeasures to limit these attacks.

Weaknesses:

- The approach takes advantage of three thresholds but the authors do not discuss how they selected them.

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Comments: This is a well executed paper showing real-world privacy implications of fitness online social networks, and in particular the fact that the location obfuscation measures that they employ are not enough to prevent a motivated attacker from learning the user's private locations, which are usually home addresses. The attack presented follows a straightforward intuition, but it's well executed. As the authors note, there are many issues with performing this attack in reality (e.g., the approximate location provided by GPS devices), and the approach convincingly takes care of them.

The countermeasures are well described, and the authors convincingly make the point that there is a tension between publishing location information and privacy. I initially was confused by which one of the proposed countermeasures was new and which ones were implementations of well-established techniques. I suggest that the authors make it more explicit, maybe by also modifying the order in which the countermeasures are described (modify radius and spatial cloaking first, fuzz EPZ intersection points later).

The generalization of the attack to other fitness social network appears to be a major selling point to me, but the experiment is currently relegated to the discussion and looks like an afterthought. I would move it to the evaluation and perhaps discuss it a bit further.

I spent most of the time when I was reading the paper wondering about ethics. The authors discuss ethical issues, but this comes very late in the paper. Moving this considerations to Section 3 would help framing the ethical context of the paper.

The approach relies on three thresholds, but the authors do not describe in detail how they determined the optimal value for them. In Section 5.1, the authors say that they took aside 10k users to establish these values, but do not provide any results on their experiments. Later in the paragraph, they say that these parameters allow them to identify 96.6% of 2.5M users, but I think that they should instead show how changing different values for the thresholds would affect detection over the 10k ground truth users (otherwise there is no point in setting them aside).

The authors consider users who set EPZs as endpoints of their runs. This makes sense, since virtually all users will want to hide their home or office location and will either start activities from there or end there. I was wondering though, what would happen if an EPZ was in the middle of an activity? Could the deanonymization process be adapted to take this scenario into account?

Nits:

- What is a virtual ride?
- In section 3.1, "arousing suspicion" should be "rising suspicion"
- In section 4, "real-world usage" would be better than "naturalistic usage"
- In section 5.1, I assume it is t_i (and not t_d as it is currently written) to be 0.1 meters?
- In section 6.4, it should be "against 0.125 mile EPZs, our effectiveness ..."

Example reaction paper/review #2

Paper: Kumar et al., USENIX Security 2018

Overall Merit: Accept

Summary: In this paper the authors investigate the potential of doing audio squatting attacks on the Amazon Alexa skill store. The authors utilize existing datasets containing speech samples from people across the US to find which words are getting confused by Amazon's Alexa. They present statistics about the confusable words and possible reasons why the confusion is happening, e.g., due to homophone pairs or due to phonetic spelling. Using their findings, the authors show that they can squat skills containing those words by registering the right pairs of skills on Amazon's Alexa. To increase the coverage of their attacks, the authors present a model for squatting based on the phonetic spelling of words and use it to find an additional 3K unique words that can be squatted. They investigate suspicious pairs of skills in the Alexa store and then show that it is possible to conduct spear-squatting attacks by identifying and taking advantage of words that are confusable between men and women, as well as words that are confusable based on the user's demographic.

Strengths:

- First study of squatting attacks for voice-controlled IoT devices
- Useful model for predicting confusions in the absence of a large corpus of spoken words
- Results are impactful

Weaknesses:

- The paper could be better if the authors had tried to conduct these attacks against popular skills

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Comments: This is a great paper that highlights issues that voice assistants have and which we need to address as we rely more and more on voice-controlled systems. I appreciated the blend of linguistics and computer security which demonstrates that we need to collaborate across different research areas as we are incorporating IoT devices into our physical environments. The experiment that is missing (and I understand the complications for making it happen) is to attempt to measure the "squattability" of existing popular skills. Squatting skills about cat facts and breathing is substantially less catastrophic compared to squatting skills about ride-hailing and banking. The authors could have applied their algorithm for predicting confusion based on the phonemes against the N most popular skills and report on how many are "theoretically" squattable. That, and perhaps a small-scale experiment with a few users, could be a precursor of a larger follow-up study involving crowd-sourcing and human subjects.

The authors should compare their work with the paper titled "Soundsquatting: Uncovering the use of homophones in domain squatting" by Nikiforakis et al. The ideas of that paper are very similar to the core ideas of this paper but they are applied to a different domain (namely domain names, instead of Alexa skills on Amazon). Moreover, the authors discuss the recent work on combosquatting but they are not citing the appropriate paper (they also seem to be using the first author's first name instead of his last name).

Defense-wise, a possibility that is worth exploring is the use of probabilities for word transitions. For example, using a real-world text corpus, one can find that the probability of the word "facts" following the word "cat" is significantly higher than that of the word "fax". These probabilities can be incorporated into the transcription process to at least protect against skill squatting where multiple words are required. This is in fact how various string segmentation algorithms work (Chapter 14 of the book titled "Beautiful Data: The Stories Behind Elegant Data Solutions" discusses such an algorithm). Nits: N/A

The “Detailed Comments” Section

- I wrote up some tips on reviewing [here](#)
- If I update the tips, I’ll let you know

Discussion questions

- In every review you write, write down 1-3 discussion questions
- These should be the burning questions you have after reading the paper, that you wish you could ask the authors
- You (audience/person who asked the question) may be called on to lead a mini-discussion based on your discussion questions
- This has been an excellent source of inspiration for in-class discussion in the past, so really think about these

4 - Present literature

- Majority of class time will be student presentations and discussions on papers
- One-Two discussion leaders/presenters per class
- Usually two papers per class, to enable more in-depth discussions

4 (cont) Responsibilities of the presenter

- Create a ~20 minute conference-type presentation on the topic to be discussed
- Briefly summarize the key idea of the work (~5 minutes)
- Identify 1-3 related citations from references in the assigned reading, outside knowledge, or your own literature search
- Use your summary of the paper and explanation of related citations as a jumping off point for discussion
- **Go through class review submissions and organize a discussion around the discussion Qs/other confusions brought up by your peers**
- You will be evaluated primarily on the quality of discussion, and you must incorporate additional citations into your presentation

Getting feedback on slides

- For presenter: If you prepare your slides early, Chris/Kris/Hansung are happy to give you feedback on them
- More info on the timeline in a few slides

4 (cont) Responsibilities of the presenter

- Requires the technical preparation necessary for writing a review, but also much more!
- Audience engagement is vital
 - Construct a narrative
 - Engage the audience
 - Identify an insight
 - Argue a point
 - Extend an argument
 - *What surprised you/made you look at the world a little bit differently?*
- Relate what you've learned, and what strikes you about the work: be engaged with the content

4 (cont) Responsibilities of the presenter

- Don't regurgitate conference presentations, don't reuse assets from the authors' slide deck... the goals of your talk are different
- Pace the talk so that you're not rushing or dragging
- Think about the goals of your presentation
- Leave audience with the high points in their head
- Practice and prepare!
- Read <http://pages.cs.wisc.edu/~markhill/conference-talk.html>

Weekly logistics / summary

- 1-2 papers to read per class.
- Each paper presented by someone in class
- For audience (non-presenter)
 - Review due dates
 - For Wednesday class: review due Sunday (AoE) before
 - This gives presenter two days to organize discussion questions
 - Reviews will be shared immediately to whole class
- For presenter
 - If you want instructor feedback on slides, send slides by or before the paper deadline. Send 2+ days before → We promise feedback.
- For everyone: you can “skip” 2 paper reviews before it starts to negatively impact your grade

For first several weeks

- We need to fill the paper pipeline
- Staff and class: Derive paper reading list.
- Tell me your conflicts
- We will post the paper list
- Tell us your preferences

5 - Term project

- The course project requires the students execute some original research in architecture
 - Demonstrate applied knowledge
 - Don't try to learn some new (non-architecture) field
 - Be realistic about what is possible in a one semester
 - However, the work should reflect real thought and effort
 - The grade will be based on: novelty, depth, correctness, clarity of presentation, and effort.
 - ??? students per group; single person suggested if you want to work in security
 - Details on project selection + progress reports to follow

5 - Term project

- The chief product of the project will be a 10-15 page conference style paper. There will be several milestones:
 - Project Ideas
 - Project Choice (~Week 4)
 - Abstract, Background and Related Work (~Week 6)
 - Experiment Proposal (~Week 8-9)
 - Project Status Slides (Week 10: late October)
 - Project Presentation (~Week 16)
 - Final Project Write-up (TBD, Finals Week)
- This is a very important factor in your grade!

Grading

- See website

Let's go around the room

- 1 minute per person
- Who are you
- Your academic background
- Your goal through taking this course
- Something funny about yourself